Roofline Lighting in Rev Render

UI/UX handoff for the dev team

Scope

This document specifies the screens, controls, data and behavior of the new **Roofline Lighting** mode inside the existing Rev Render flow. It covers UI/UX only. How a render request is assembled and sent to the image model is **out of scope** and handled separately; this document only defines what the UI collects and hands off.

A live clickable mockup accompanies this document. Sample data in it (Tampa job location) is simulated. The mockup also contains Results and Send screens; these are outside this spec, which ends at Generate Renders.

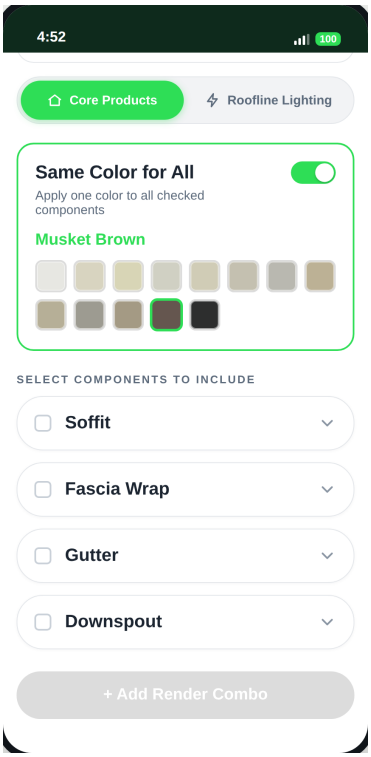
Core principles

- **Core Products stays exactly as it is today.** No visual or behavioral change to the existing Configure Render screen.
- **A render is Core OR Lighting, never both.** A switch under the photo selects the mode.
- **The rep picks a look, nothing else.** No components, no cover type, no channel color, no time of day, no Lights Off. Channel always matches the home's existing trim; scene is always dusk.
- **Everything is one-tap or hold.** Favorites and Recently Rendered sit on top for repeat work.
- **Batch friendly.** Multiple looks can be added at once; each becomes its own render in the queue.

Screen flow

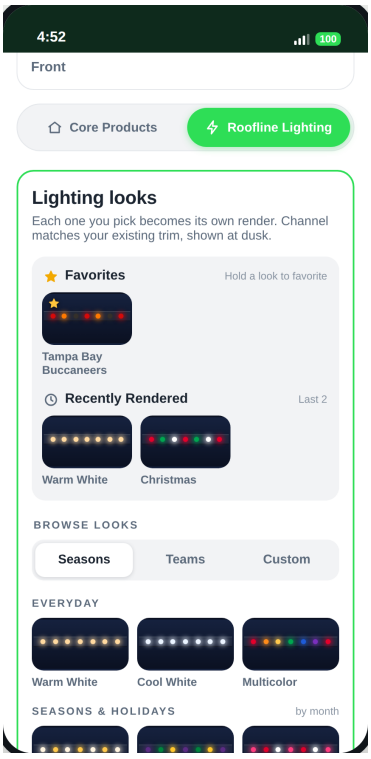
#	Screen	Purpose
1	Configure Render (mode = Core)	Unchanged. Existing screen.
2	Configure Render (mode = Lighting)	Pick looks: Quick picks, then Seasons / Teams / Themes / Custom.
3	League browser sheet	Full team list by league / conference.
4	Custom look builder sheet	Build a one-off look with a color wheel.
5	Queue	Review renders (same list as Core, lighting rows show Channel and Lights).

1. Mode switch and unchanged Core screen



Core Products (unchanged)

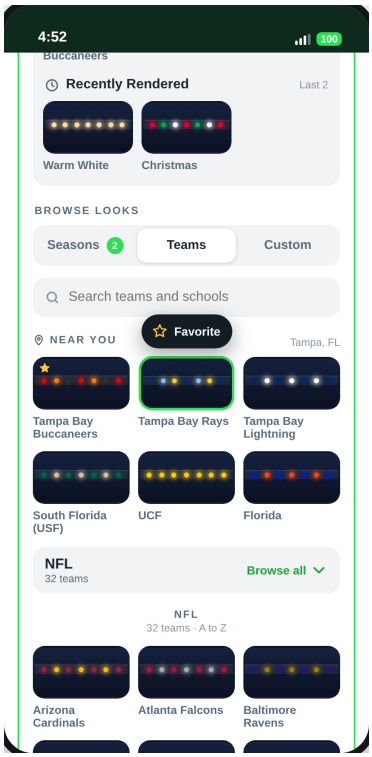
- Reference only. Ship exactly as it exists today.
- A two-option segmented control sits directly under the photo: **Core Products | Roofline Lighting**. Core is the default.
- Switching modes swaps only the configuration area below the switch. The photo, header, and bottom action area stay in place.
- Switching modes with items already in the queue: the queue keeps its rows. Do not mix modes inside a single render; each queue row is tagged with its own mode.
- **Open:** if the rep has an unsaved selection in one mode and switches, keep it in memory per mode (mockup does this) and restore on return.



2. Lighting screen structure (top to bottom)

- **Quick picks** panel: Favorites (star) and Recently Rendered.
- **Browse Looks** label, then four tabs with count badges: **Seasons | Teams | Themes | Custom**. Count = number of selected looks in that tab.
- Tab content (tiles or team list).
- **Renders selected** box: its own card listing every selected look across all tabs, with count.
- Summary line: "N looks - matches existing trim".
- Primary button: **+ Add Render Combo** (1) or **+ Add N Render Combos** (N). Disabled at 0 selected.

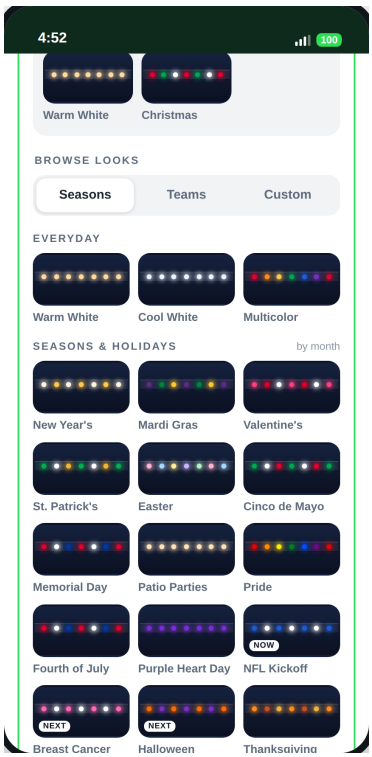
3. Quick picks



Favorites and Recently Rendered

- **Favorites:** no limit. Shown as tiles, newest favorite first.
- **Recently Rendered:** the last **3** looks the rep rendered, excluding anything already a favorite. Most recent first.
- Tapping a tile toggles selection (same as any other tile). Selected state must match across every place the same look appears (Quick picks, its tab, selected box).
- **Add to Favorites:** press and hold a tile for **450 ms**. Trigger a light haptic tap, show a small popover "Added to Favorites" (or "Removed"). Cancel the hold if the finger moves more than 10 px or the list scrolls.
- Also provide a visible star toggle on each row in the Renders selected box. This is the discoverable path and the accessibility path (long-press alone is not accessible).
- Hint text under the section: "Hold a look to favorite".
- **Haptics:** browsers on iPhone cannot vibrate, so the mockup only simulates it. Native build should use the platform's light impact feedback on both iOS and Android.
- **Open:** record Recently Rendered when the render is *generated* (label says rendered). The mockup records at Add; dev should confirm and move to Generate.

4. Seasons tab



Everyday and holidays

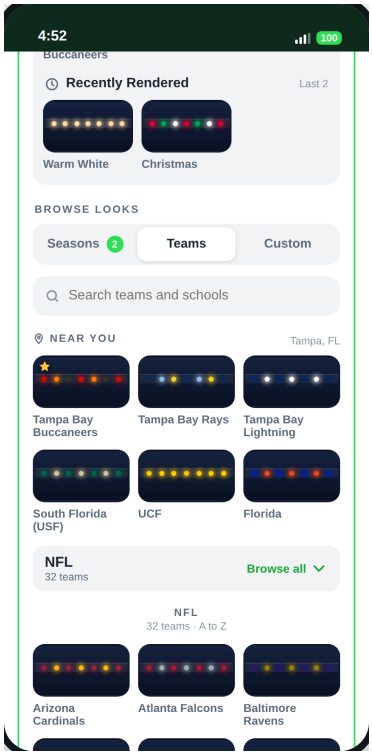
- **Everyday** group (3): Warm White, Cool White, Multicolor.
- **Holidays** group (18), sorted by calendar month then name. Each tile shows a mini light strip in the look's colors and the name.
- Jan: New Year's. Feb: Mardi Gras, Valentine's. Mar: St. Patrick's. Apr: Easter. May: Cinco de Mayo, Memorial Day. Jun: Patio Parties, Pride. Jul: Fourth of July. Aug: Purple Heart Day. Sep: NFL Kickoff. Oct: Breast Cancer Awareness, Halloween. Nov: Thanksgiving, Veterans Day. Dec: Christmas, Hanukkah.
- "Now" and "Next" tags mark the current and upcoming month's holidays, computed from the device date. Nothing is preselected.
- Presets are not editable. Custom needs go to the Custom tab.

4b. Themes tab

A fourth tab, **Themes**, holds non-sport, non-holiday looks in four categories: **Marvel** (15), **DC** (13), **Military** (6 branches: Air Force, Army, Coast Guard, Marine Corps, Navy, Space Force) and **Countries** (34 flags, for example Cuba, Mexico, Puerto Rico, Colombia, Brazil, Ireland, Italy, UK, Canada). It uses the same pattern as Teams: a selector button showing the current category and count with "Browse all", which opens a bottom sheet grouped Heroes / Armed forces / World. The grid below shows the chosen category A to Z (default Marvel). Each look is a light palette drawn from costume, service or flag colors; no character art, flags or logos appear in the UI. Colors that are too dark to glow (such as black on the German flag) are dropped automatically. Tiles select, favorite and queue like every other look. No search on this tab; add one if categories grow. New categories are added as data only (catalog group = theme, category = name), with one row in the category list. Sheet grouping is defined in config.

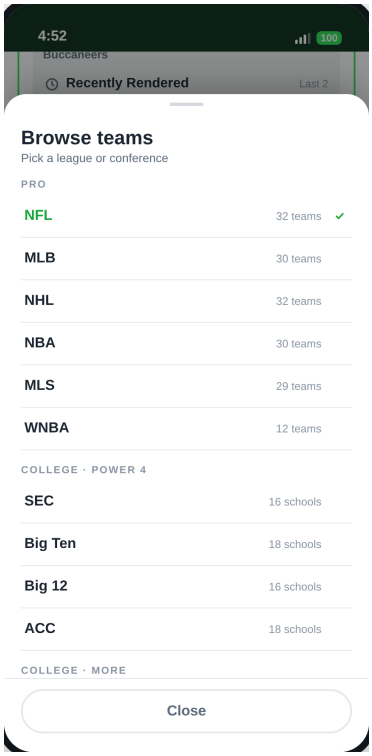
Open (Legal): Marvel and DC character names may require a license or trademark review; fallback is generic names (for example, Web Slinger, Caped Crusader). Military branch names and insignia rules should also be checked (names and colors only, no insignia). Flags are generally safe; some may be politically sensitive for certain customers, so let the business decide the final country list.

5. Teams tab



Search, Near you, Browse

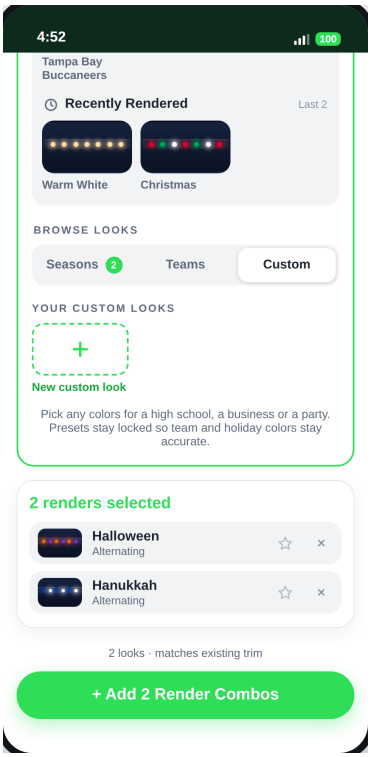
- **Search field** at the top. Matches team name, city, nickname, and school, ignoring case, punctuation, and accents ("virginia" returns Virginia, Virginia Tech, West Virginia). The input keeps focus while results filter live.
- **Near you** row: up to 6 tiles based on job location. Sourcing rules are in section 10.
- **Browse** button opens the league sheet. Below it, all teams A to Z in one list, with league tags.
- Tiles show a mini strip in the team's colors and the team name. Same selected/favorite behavior as Seasons.
- Categories (268 teams): NFL 32, MLB 30, NHL 32, NBA 30, MLS 29, WNBA 12, SEC 16, Big Ten 18, Big 12 16, ACC 18 (includes Notre Dame), Other FBS 20, HBCUs 15.



League browser sheet

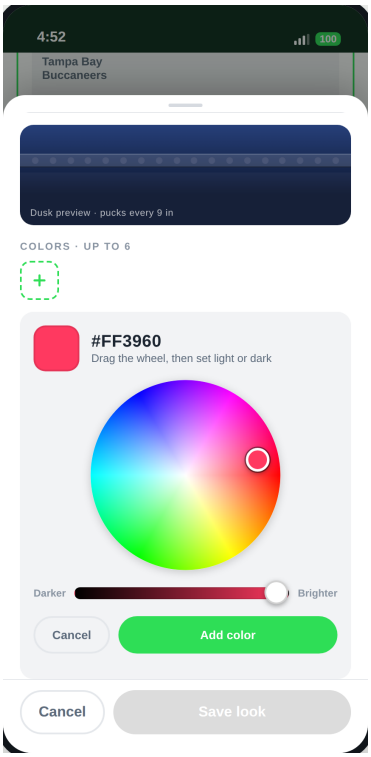
- Bottom sheet listing each league / conference with team count.
- Choosing one shows that group's teams alphabetically, with the count of selected teams in that group.
- The earlier horizontal scrolling rail was replaced by this sheet because it wobbled and overflowed on small phones.

6. Custom tab and builder



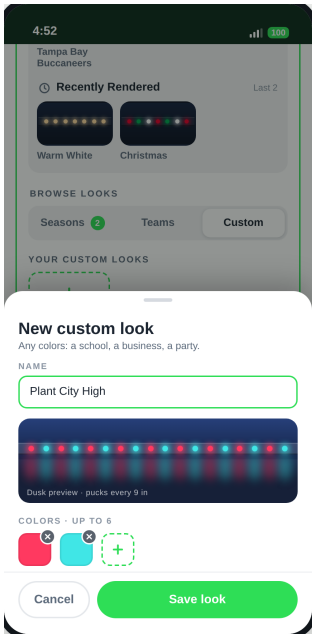
Custom tab

- Starts with **nothing selected** and no custom looks.
- A "+ New custom look" tile opens the builder sheet.
- Saved custom looks appear as tiles and behave like any other look (select, favorite, appear in the selected box).
- Custom looks are the only place the rep can invent a look. There is no "not listed? make a custom look" prompt on other tabs.



Builder sheet

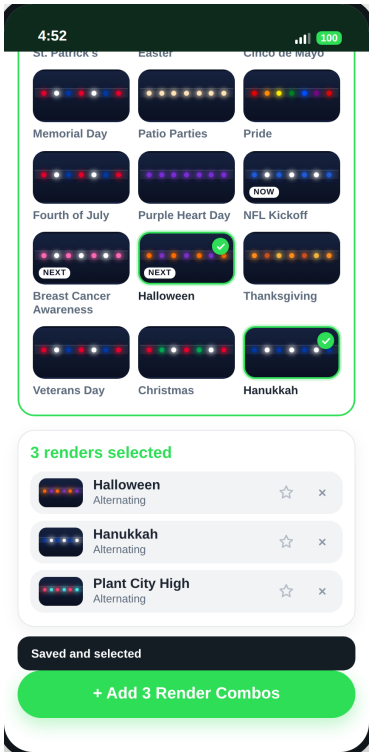
- Fields: Name (required), up to 4 colors.
- **Color picker:** a color wheel (hue = angle, saturation = distance from center, center is white) plus a brightness slider for lights and darks. Dragging updates the swatch live.
- **No pattern control.** One color renders as a single color; two or more render as alternating.
- **Dark colors:** a color that is too dark to glow (brightest channel under 40 of 255) is dropped from the look, with a small inline notice. LEDs cannot emit black.
- Save is disabled until there is a name and at least one usable color.
- Add / remove color swatches beneath the wheel.



Builder with colors added

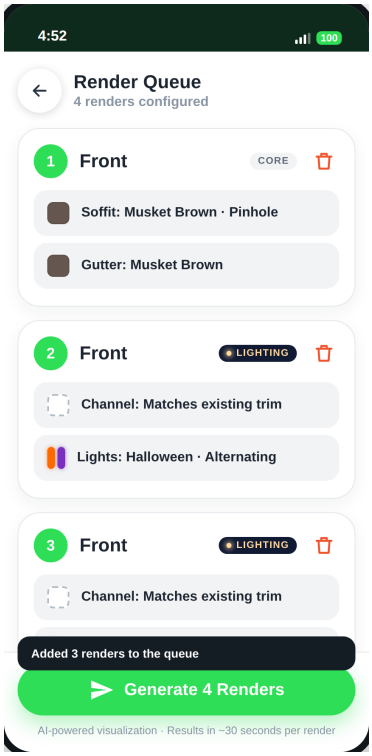
- Selected swatches shown in a row; tap one to edit it on the wheel; tap x to remove.

7. Selection and queue



Renders selected box and Add

- Separate card under the tabs. Header: "N renders selected".
- Each row: mini light strip, look name, pattern (Single color / Alternating), star (favorite toggle), and x (remove). Rows are not editable.
- The count spans all tabs. Deselecting anywhere updates everything.
- The Add button turns each selected look into one queue item and clears the selection.



Queue rows for lighting

- Same queue component as Core. A lighting row shows the photo thumbnail plus two lines:
- **Channel:** Matches existing trim (dashed square as the color chip, since no color is chosen).
- **Lights:** look name, then Single color or Alternating.
- Removing a row returns nothing to the selection. Generate Renders is the last step in this spec; everything after it is out of scope.

8. UI data model and hand-off payload

Field	Type	Notes
mode	"core" | "lighting"	Per photo.
sel	string[] of lookIds	Current multi-selection across tabs.
fav	string[] of lookIds	Unlimited. Persisted (see open decisions).
recent	string[] (max 3)	Most recent first; excludes favorites at display time.
customs	Look[]	User-created looks: id, name, colors[].
tab	seasons | teams | custom	Last active tab restored.
cat	string	Selected league in the browse sheet.
q	string	Team search text.

Look record (catalog item)

Field	Description
id	Stable slug, e.g. nfl-tampa-bay-buccaneers, halloween.
group	everyday | holiday | team | custom
category	Everyday, month, or league / conference.
name	Display name.
colors	Array of hex strings, 1 to 4, light colors as they should appear when lit.
pattern	single (1 color) or alternating (2+). Derived, not user-set.

Per-render output from the UI

For each queue item, the UI passes the following to the render service. Nothing else is collected.

```
{ photoId,
  mode: "lighting",
  lookId, lookName,
  colors: ["#RRGGBB", ...],
  pattern: "single" | "alternating",
  timeOfDay: "dusk" (constant),
  channel: "match_existing_trim" (constant),
  coverage: "all visible roof runs" (constant) }
```

How this is turned into a model request is out of scope here. A full machine-readable catalog of all 289 looks (id, group, category, month, name, colors, pattern) is provided as a companion CSV.

9. What changed versus the earlier vendor lighting spec

Earlier spec	This design
Component picker (gutters, soffit, fascia, downspouts)	Removed. Whole roofline; channel matches existing trim.
Cover / channel type and channel color	Removed. Always matches trim.
Same Color for All, per-component colors	Removed.
Lights Off option	Removed. Always lit.
Time of day selector	Removed. Always dusk.
Editable presets	Removed. Presets are fixed; custom looks live in the Custom tab.

Earlier spec	This design
Holiday presets only	Added Everyday, Teams (268), Themes (68: Marvel, DC, Military, Countries), Custom, Favorites, Recently Rendered, Near you, batch add.

10. "Near you": how nearby teams are chosen

Recommendation: radius first, same-state backfill. State-only matching fails at borders (Kansas City, Cincinnati, Philadelphia) and inside large states (a Miami job would rank Jacksonville and Tampa teams the same as Miami's). Radius handles metros; state fills gaps in rural areas and covers college teams, whose fans are statewide.

Location source, in order

- **1. Job / homeowner address**, geocoded to lat/lng (use the geocoder Rev Render already calls for the address).
- **2. Rep device location**, only if permission is already granted. Never prompt just for this.
- **3. Rep's office / market** from the account profile.
- **4. None available:** hide the Near you row. No fake defaults.

Data pulled

Ship a static **team location table** with the looks catalog: one row per team with lookId, league, home venue or campus lat/lng, and state. No live API call and no third party at runtime. Distance is a haversine calculation on device (or in the same service that serves the catalog). Refresh the table with the yearly catalog update (relocations, expansion teams, conference moves).

Ranking algorithm

Step	Rule
1. Pro teams	Nearest team per league (NFL, MLB, NHL, NBA, MLS, WNBA) within **75 mi**. In multi-team markets (NY, LA, Chicago, Bay Area) allow up to 2 per league. Cap about 3 tiles.
2. College teams	Schools in the job's state, any tier, ranked by distance. If none, schools within **150 mi**. Cap about 3 tiles.
3. Fill to 6	Unused pro slots go to college and vice versa.
4. Backfill	If fewer than 6, add same-state teams by distance, then nearest teams overall. Relabel the row "Nearest to you" when nothing is in range so it is never empty or misleading.
5. Order	Pro first, then college, each by distance ascending. Deduplicate against Favorites only if the row would otherwise be empty.

Tunables (config, not hard-coded): PRO_RADIUS = 75 mi, COLLEGE_RADIUS = 150 mi, SLOTS = 6, MAX_PER_LEAGUE = 2.

Example: Tampa, FL

Tile	Approx. distance	Why
Tampa Bay Buccaneers	~0-5 mi	NFL within radius
Tampa Bay Rays	~20 mi	MLB within radius
Tampa Bay Lightning	~0-5 mi	NHL within radius
South Florida (USF)	~0-10 mi	In-state, nearest college
UCF	~85 mi	In-state, next by distance
Florida	~125 mi	In-state, next by distance

Privacy

- Compute locally against the bundled table. Do not send the homeowner's address anywhere new for this feature.
- Store only the resulting team ids or a city label, not coordinates, with the job.

- If the address changes, recompute.

11. Rules and edge cases

- Selection persists when switching tabs and when the browse sheet opens or closes. It clears after Add.
- A look can be selected from Quick picks, Seasons, Teams or Custom; the selected box de-duplicates by lookId.
- Add is disabled at zero selections. Adding N looks creates N queue items, each with the same photo.
- Search has no results: show "No teams match" and keep the input focused.
- Long-press must not fire on scroll. Cancel when movement exceeds 10 px or a scroll event occurs.
- Text on colored strips must remain legible in dark mode; strips carry a thin outline so white and dark colors are visible on either background.
- Every tile has a minimum 44 pt tap target. Every icon-only control has an accessible label (star, remove, tab, color swatch).
- Color wheel is not keyboard or screen-reader friendly by itself. Provide a hex input field in the builder as the accessible alternative.
- Tab counts and the selected box must always agree.

12. Acceptance criteria

- Core Products screen is pixel-identical to production with the mode switch added.
- A render can never contain both Core and Lighting items; the switch and queue enforce this.
- Lighting screen never shows component, color, cover, time-of-day or Lights Off controls.
- Favorites are unlimited; Recently Rendered shows exactly the last 3 non-favorites.
- Hold 450 ms adds or removes a favorite with haptic; scrolling never triggers it.
- Teams: 268 teams present, search finds "Virginia" and "Notre Dame", league counts match section 5.
- Near you follows section 10 and never shows an empty row when any location is available.
- Custom starts empty; wheel and brightness slider work by touch; dark colors dropped with notice.
- Queue rows for lighting show Channel: Matches existing trim and Lights: name + Single color / Alternating.
- Test devices (Android): Moto E 2022, Moto G Play 2024, Galaxy A15 5G. Plus a current iPhone. Check wheel drag performance, long-press, and sheet scroll on the small screens.

13. Open decisions and things to verify

Item	Question / risk	Owner
Persistence	Favorites, Recents and custom looks: per rep, per office, or per account? Sync across devices?	Product
Recents timing	Record at Generate (recommended) or at Add?	Product
Team colors	Hex values were entered from general knowledge and must be verified against each club's brand guide before release.	Design
Utah Jazz	Palette changed in the 2024 rebrand; confirm which set to ship.	Design
Hero names / IP	Marvel and DC character names and palettes need trademark / license review.	Legal
Theme colors	Hero palettes chosen by eye from costumes; flag and service colors from general knowledge. Verify with design.	Design
Country list	34 launch countries; confirm the list with sales / regional teams.	Product
Teams left out	San Diego FC, Golden State Valkyries, Portland Fire, Toronto Tempo (colors unconfirmed). Add when verified.	Design
Notre Dame	Listed under ACC (non-football member). Confirm with sales that reps expect it there.	Product
Conference alignment	Verify for the current season; realignment is ongoing.	Design
Haptics	Native implementation for iOS and Android; browser mockup cannot demonstrate it on iPhone.	Dev
Dark-color rule	Threshold (max channel under 40) is a starting point; tune with real renders.	Design

14. Mockup-only items

- Tampa as the job location and the resulting Near you set.
- Haptic feedback (vibrate) which iPhone browsers ignore.
- Queue thumbnails and the photo.